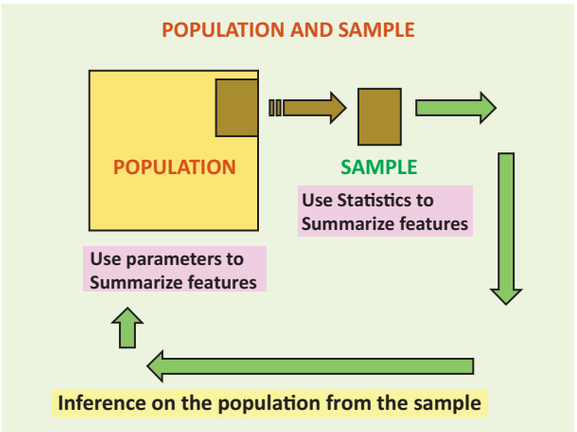
Introduction

In XI Standard classes, we concentrated on collection, presentation and analysis of data along with calculation of various measures of central tendency and measures of dispersion. These kinds of describing the data are popularly known as **descriptive statistics**. Now, we need to understand another dimension of statistical data analysis, which is called **inferential statistics**. Various concepts and methods related to this dimension will be discussed in the first four Chapters of this volume. Inferential Statistics may be described as follows from the statistical point of view:

One of the main objectives of any scientific investigation or any survey is to find out the unknown facts or characteristics of the population under consideration. It is practically not feasible to examine the entire population, since it will increase the time and cost involved. But one may examine a part of it, called **sample**. On the basis of this limited information, one can make decisions or draw inferences on the unknown facts or characteristics of the population.



Thus, inferential statistics refers to a collection of statistical methods in which random samples are used to draw valid inferences or to make decisions in terms of probabilistic statements about the population under study.

Before going to study in detail about Inferential Statistics, we need to understand some of the important terms and definitions related to this topic.

1.1 PARAMETER AND STATISTIC

A **population**, as described in Section 2.4 in XI Standard text book, is a collection of units/objects/numbers under study, whose elements can be considered as the values of a random variable, say, X. As mentioned in Section 9.3 in XI Standard text book, there will be a probability distribution associated with X.

Parameter: Generally, **parameter** is a quantitative characteristic, which indexes/identifies the respective distribution. In many cases, statistical quantitative characteristics calculated based on all the units in the population are the respective parameters. For example, population mean, population standard deviation, population proportion are parameters for some distributions.

Recall: The unknown constants which appear in the *probability density function or probability mass function* of the random variable X, are also called **parameters** of the corresponding distribution/population.

The parameters are commonly denoted by Greek letters. In Statistical Inference, some or all the parameters of a population are assumed to be unknown.

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